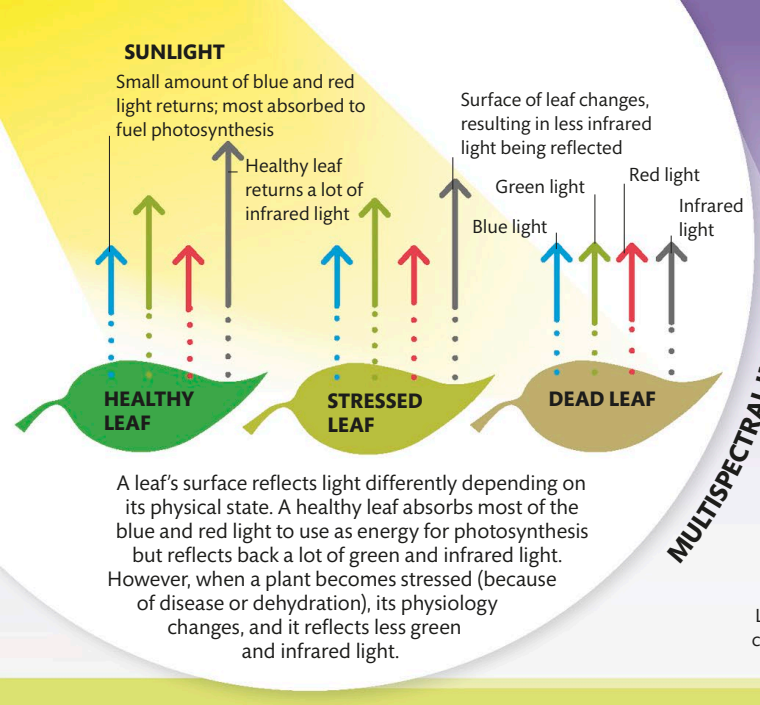
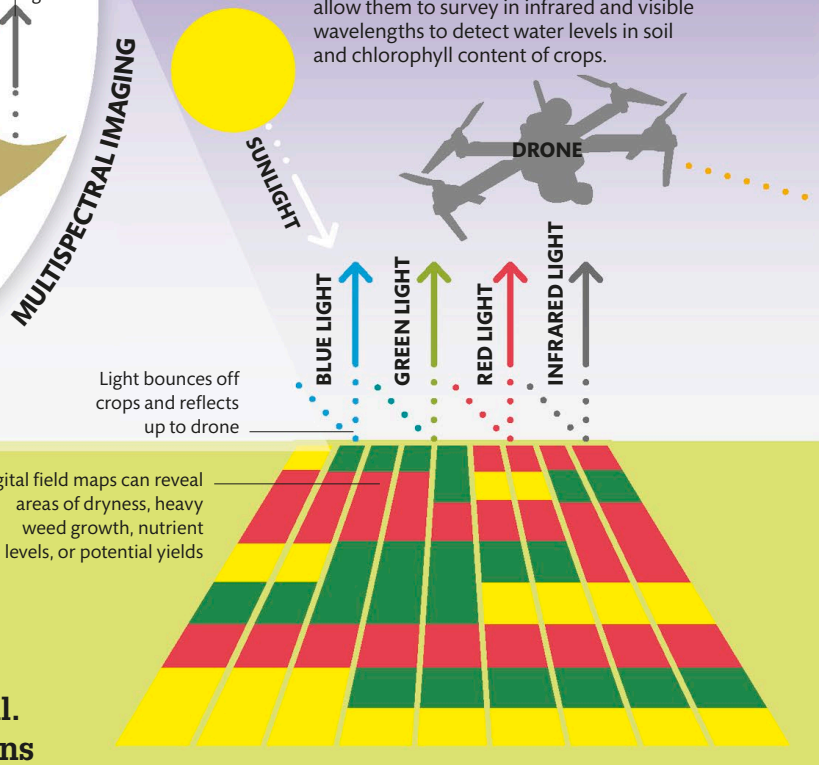


A leaf's surface reflects light differently depending on its physical state. A healthy leaf absorbs most of the blue and red light to use as energy for photosynthesis but reflects back a lot of green and infrared light. However, when a plant becomes stressed (because of disease or dehydration), its physiology changes, and it reflects less green and infrared light.



1 Remote imaging
Drones are used for aerial surveys of the land. Many drones use multispectral cameras, which have multiple lenses. These allow them to survey in infrared and visible wavelengths to detect water levels in soil and chlorophyll content of crops.

Precision agriculture

Farming is increasingly going digital. Farmers can now use communications and computer technology to collect data from their crops and animals, which they can then use to manage their farms more efficiently and remotely control their machinery.

Monitoring crops

Precision agriculture enables farmers to use data from various sources, from sensors in a field to drones and satellites, to improve crop yields and reduce waste. GPS data allows exact locations to be calculated so that each part of the field can be managed precisely. Farmers can download information about a specific site on a field, such as weed distribution or pH levels in the soil, and treat each location individually. Agricultural equipment connected to the Internet allows a farmer to manage his or her farm remotely.

MONITORING LIVESTOCK

Livestock can have various sensors attached to them that provide useful information for farmers. Chips and tags enable tracking, useful for farmers searching for lost animals, or allow precise identification of animals through regulatory and retail systems. Sensors can also alert the farmer to any medical problems or indicate whether an animal is ready to mate or give birth.

